



The reliability of character recognition: An Australian & New Zealand expert-novice comparison study in the interpretation of chemically recovered serial numbers

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ABSTRACT

The chemical recovery of a defaced serial number is a common forensic science practice, however it is not understood how proficient experts perform in correctly identifying recovered serial numbers. Understanding the accuracy of experts and how they compare to novices in character recognition can help to establish a baseline for this expertise. In this study an expert-novice comparison assessment was completed involving 118 test plates, each stamped with six randomised alphanumeric characters. The plates were defaced and chemically recovered before being viewed by multiple participants over six time intervals. A total of 3169 character inspections were completed. An assessment of confidence and error rates were calculated for both expert (trained) and novice (untrained) participants. Errors were counted when a participant interpreted a different character to that of the ground truth and believed the result was accurate for reporting. The results showed a similar (2.3 % and 2.4 %) error rate for the cohorts, however a statistical difference in confidence levels was recorded, demonstrating the more conservative nature of experts. This study aims to assist in validating practitioner interpretations, through addressing some forensic science criticisms, such as establishing error rates to routine scientific practices.

1. Introduction

A serial number, an identifiable code applied to an item often at the time of its creation, can be used to track the item through its manufacture, stocktaking, sale and distributions. The marking is used to regulate products, such as firearms and vehicles [1]. Criminal entities may alter or remove serial numbers in an attempt to prevent tracing by law enforcement [2]. If seized by police these items may be sent for forensic examination to identify a chain of ownership and potential offender. To combat any removal efforts, forensic practitioners will use laboratory-based methods such as chemical etching, to recover defaced markings.

The application of recovering a defaced serial number (often referred to as *serial number restoration*) is a practice that highlights deformations within a materials structure [3]. The process can be successful using different surface media, (including metal, plastic, wood, and leather) [3, 4], and although the treatments for each vary, they rely on the same principle: Each method attempts to provide a visible distinction of the

deformation that is produced in the material during the application of the serial number.

Recoveries are heavily dependent on the substrate and the application method. Cold stamping characters into a metal creates a permanent cavity mirroring the shape of the die stamp, and an area of plastic deformation that continues to extend below the visible indentation of the character [5,6]. Thus, there are circumstances where even if the character indentation is removed, the disruption to the crystalline structure below the indentation retains detail of the character stamping that may be recovered [3]. A controlled chemical etching of the surface can visibly highlight the difference between the deformed and undeformed areas of the metallic structure [2,3,7,8]. Angled light is used to create shadows from the surface contours to assist the interpretation of results. This process is illustrated in Fig. 1.

There are a number of factors which can influence the likelihood of a decipherable recovery, such as the stamping depth, obliteration depth, impression type, character type, etc. [4,9,10–12]. Furthermore, recent studies have given weight to specific formulations being more effective

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based on the chemical properties of the material; however Fry's Reagent (or close variants of) has repeatedly shown to be effective across a range of metal properties [13].

Although there have been many published papers on serial number restorations, the focus is often centralised around testing reagents and not the reliability of practitioners' interpretations, efficiency or accuracy. For example, sample plates in these studies were often marked with the same numeric or alphanumeric characters [9,14–18]. Although restorations are regularly practised in forensic laboratories there appears to be no published empirical study which assesses the reliability of interpretations.

1.1. Interpretation

The recognition of alphanumeric characters is a skill developed during childhood and continues to develop as we age [19]. The ability later becomes a subconscious practice due to its frequency of use. A previous publication by the authors theorises the interpretation process, which has been described as the following stages: Distinguishing the character(s) from background noise; Character isolation/segregation; Feature extraction of the character; Distinguishing relative positions of features (zoning); Application of context; and Classification [20].

As humans our actions inherently involve a degree of error. The interpretation of alphanumeric characters is no exception. Although the degree of error is likely context based, one previous study has stated that the human eye has an approximately 4 % error when reading without context [21]. By design of modern English characters alone, it would be expected that specific characters would incur higher levels of misinterpretations. For example, the number 0 could be easily mistaken as to the letter O, but easily distinguishable from the number 4 [20]. Although some studies have broached alphanumeric character interpretations of recovered markings, none appear to have assessed participant accuracy.

A 2016 serial number restoration study by Fortini (2016) touched on

analysing the 'judgements' of male to female participants across different age groups, however this was focused as an assessment of the reagent rather than the individual. Fortini does state that the 'observers' were not aware of the original number [13], however no indication was provided as to observer accuracy, nor the information the observers were exposed to, such as serial number length, format, font, etc. Another study by Peacock (2022) involved collecting and comparing data on the correctly identified recovered characters by examiners. Although this paper was also centralised around the assessment of reagents, the paper does comment that anecdotally operator experience (in serial number restoration) is likely to play a role in determining success [22].

As interpreter accuracy has not been thoroughly assessed, this begs question, are the characters identified by the individuals really the characters that were previously removed? Further, as interpreter data has either not previously been collected or published, it is unknown if participants trained in serial number restorations, due to their acquired knowledge of subsurface crystalline structure deformation resulting from the stamping process, are more reliable at interpreting recovered markings compared to the layperson. There also appears to be no previous research which assesses the frequency of errors in serial number restorations, their origins, or any mitigation strategies [20].

This paper aims to assist in validating the interpretations of chemically recovered serial numbers. It is hoped that by comparing trained and untrained participants, it can be determined if training allows for a more accurate interpretation of results, and how it is reflected on participant confidence. This paper addresses the visual interpretation aspect of recovered characters only, and thus it is not a complete assessment of chemical restoration as a whole process. The experiment design aimed to align with other expert-novice comparison studies, e.g. forensic fingerprints [23], whereby only the interpretation/identification aspect of the practice is assessed by both trained and untrained participants. Through combining research into the various aspects of serial number restoration a more holistic assessment of the practice may be obtained.

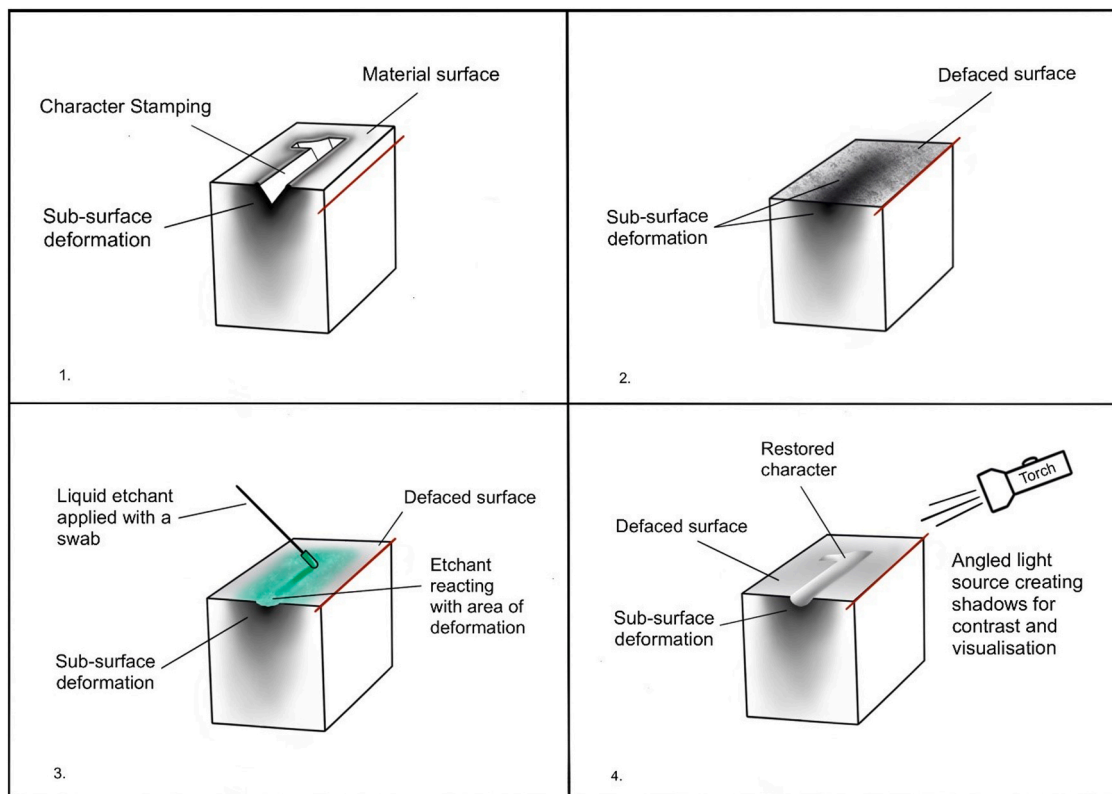


Fig. 1. Chemical etching character recovery process.

2. Materials and methods

2.1. Plate formation

Two materials were assessed as part of this study, due to their prevalence of the manufacture in firearms, 4140 medium carbon steel and '300 Plus' mild carbon steel. The metals were cut from a bar using a circular cutter to create test plates at approximately 50 mm height, 50 mm width, and 7 mm depth. One face of each plate was smoothed to a relatively uniform surface finish using a belt sander with 240 grit sandpaper, and press stamped using a hydraulic press with six random alphanumeric characters, consisting of A to N, P to Z, and 0–9. (The die stamp character for the number 0 and the letter O in this set was identical, and therefore not duplicated in this study.) Randomization was accomplished using an online random alphanumeric generator, with the total number of characters used amongst the plates monitored on an Excel spreadsheet to ensure a consistent distribution. Each plate was then hand engraved with a reference marking on the opposing face of each plate. Plate thicknesses (depths) were measured at five points using a digital micrometre gauge. The thickness and reference markings were recorded for later reference and comparison.

Sharp-faced die stamps using an Arial-type font were used to press stamp all characters. To compensate for the materials difference in tensile strength, the 4140 steel plates were stamped at 300 kg/cm² (29.42 MPa), whilst the softer 300 Carbon Plus plates were stamped at 150 kg/cm² (14.7 MPa). This provided a visually similar serial number depth between the two materials. The reference numbers given to each plate were correlated against each six-character serial number using an Excel spreadsheet.

2.2. Character removal & measurements

Each plate was ground using a surface grinder until the six character serial numbers were no longer visible, and plate's thicknesses were remeasured (again at five points) with the digital micrometre. As per previous studies [16,22] this was to determine the amount of material removed, and thus provide an approximation of 'obliteration depth'. This measurement was essential to ensure participant results were not swayed by the depth of character obliteration. An approximate character stamping depth of 0.2 mm was determined, providing an 'obliteration depth' range between ~0.03 mm and ~1.00 mm.

As the plates were ground in batches (primarily 10 plates at a time), the slight differences in plate thicknesses allowed for intended variations in the depth of material removed ('obliteration depth'). The minimum amount of material removed from the plates was 0.233 mm, with a maximum of 1.167 mm. On average the amount of material removed was determined to be approximately 0.5 mm.

2.3. Participant selection

86 participants from across Australia and New Zealand were assessed in this study. Participants were divided into two test cohorts (an 'expert' group and a 'novice' group) depending on their training in serial number restoration. Both cohorts contained male and female participants of varying ethnicity, education and age. The study aimed to provide a relatively even distribution of experts to novices, however no other discretion was applied to select participants. A total of 45 'experts' and 41 'novices' were selected.

Prior to undertaking any involvement in this study, all participants were asked to complete a short background questionnaire. This asked participants to record their name and a four-digit identifier (which was later de-identified). It further asked participants if they had previous experience with serial number restorations and pattern matching studies. Ethics approval was obtained through the University of Technology Sydney (ET21–6002) to complete the study, and participants were required to complete a signed consent form for the collection of

their responses.

Participants were provided general information about the project as well as specific instructions needed to complete the practical assessment prior to undertaking any interpretations. This included information about the stamping, style and format of the serial numbers. As per standard laboratory practice a handheld LED torch was used to provide various angles of light during plate assessments, and participants requiring prescription glasses were encouraged to wear them. However to ensure each participant was provided equal access to equipment no further optical aids (e.g. incl. magnifying glasses or microscopes) were permitted. Each participant was instructed to compare the visual result produced after restoration to a reference sheet illustrating the font of each character used in the study, and draw what they could perceive of each character. Participant's interpretation of the recovered markings on each plate were then recorded on a results page.

To fall within the expert cohort, practitioners needed to be specifically trained in undertaking serial number restorations using chemical etchants. This primarily involved the participants undertaking a specific training module (involving theoretical and practical assessments) as part of their training. Due to the specific nature of the training and work, the expert group were practitioners employed within government or scientific facilities that undertake serial number recoveries.

On the other hand, participants in the novice cohort had not undergone any specific training in serial number restorations. This group consisted of individuals who often worked in close proximity to those in the expert cohort, but ranged from no knowledge of the practice to possessing a fundamental understanding of the theory and seeing the procedure followed, but not having any further experience. The novice group were generally administrative staff of government or scientific agencies, or university students with some scientific background.

3. Data collection & analysis

Each of the obliterated plates were cleaned with acetone and then chemically etched to recover the six alphanumeric characters. Restorations were completed within forensic facilities within Australia (at Sydney, Melbourne, Brisbane, Perth, Hobart) and New Zealand (Auckland) using various etchants depending on the laboratory's standard chemicals and operating procedures. The chemical etching of the plates was completed by wiping a cotton swab dipped in etchant gently across the defaced surface (the swabbing technique), under extraction fans or fume hood. A hand-held multi-phase torch was used to provide oblique light to the surface for inspections. The plates were treated with the etchants until all characters could be identified, or up until the point where the individual perceives that the markings are beginning to deteriorate, for a maximum of two hours. If no result was visible after two hours the etching process was terminated. After recovery, each plate was dabbed dry using paper towel and gently wiped with acetone to prevent further unnecessary etching.

Each etched plate was individually inspected by up to six participants (including the participants undertaking the etching process) who were unaware of the six-character serial number that had been obliterated. Participant inspections occurred at the time of restoration, 5 min after restoration, then at the 15 min mark, 30 min, 45 min and 1 h after the completion of chemical restoration. The time interval assessment aimed to determine an ideal time for recovered character assessments. Participants were asked to keep their results confidential, thus participants were unaware of previous responses.

A result sheet per plate and time interval was completed which asked participants to record their interpretation and provide a confidence rating to each character identified. When each plate had been inspected by the (up to 6) participants, the results were collected and digitised onto an excel spreadsheet, documenting each individual result and any discrepancies to the known truth (the actual stamped characters on each plate). Each line on the excel spreadsheet represented a participants interpretation, and included participants respective cohort (expert or

novice), their level of training, the time interval in which the assessment was completed (e.g. 15 min), the number of characters identified, their confidence rating and a confidence percentage. Data analysis was completed by inter-comparing each line of results on excel, using filter dropdowns and Excel formulas.

3.1. Error rate testing

Recovered characters that were misidentified by participants were used to form part of the error rate analysis in this study. A misidentification that contributes as a character error was counted when a participant records a result which differs from the known (pre-obliterated) serial number, and further commits a level of reporting confidence to that result. Each misidentification and character error found in the study was recorded and totalled so that commonalities could be identified. Fig. 2 shows a chemically etched plate with six recovered characters.

In this example, a participant identified the characters ‘H I Y R B Z’, however the actual stamped marking (ground truth) was ‘H I Y 9 3 Z’; thus, the fourth and fifth characters in the sequence have been misidentified by the participant. Of the two misidentified characters, the participant was unconvinced by the fourth character being an ‘R’, and answered accordingly. However the participant believed more firmly in their interpretation of ‘B’ being the fifth character. As the participant incorrectly reported the character as a ‘B’ this constitutes a character error and forms part of the error rate analysis.

3.2. Confidence testing

The confidence of participants was assessed in this study using a rating system. The system required participants to attribute a numerical figure to their level of confidence (one to five) for each character identified during the plate inspections. Participants were advised prior to the commencement of the inspection that one [1] was the minimum score (meaning least confident) and a score of five [5] being the maximum, or most confident. Thus for each plate, where a participant identified all six characters, the minimum score was six, and the maximum score was 30.

4. Results and discussion

4.1. Plate assessments

118 plates were assessed as part of this study. 59 plates were constructed of 4140 Steel (29 polished, 30 unpolished), and 59 plates constructed of 300 Plus steel (30 polished, 29 unpolished). The study was completed in operational laboratories around regular duties, not all 118 plates could be assessed at each of the six desired time intervals. All 118 plates were assessed at the time of recovery (time 0), 114 were assessed further at 5 min, 110 at 15 min, 110 at 30 min, 97 at 45 min and 96 at the 60 min mark. This provided a grand total of 645 plate inspections completed. Table 1 below shows the number of character inspections completed between the expert and novice cohorts.

By the end of the plate assessments, the expert cohort contributed to



Fig. 2. Ambiguous recovered serial number.

Table 1
Number of characters identified.

Time	Plates Assessed at Time	Total Potential Characters Assessed	No. Characters Identified	% Assessed by Trained (Expert) Cohort	% Assessed by Untrained (Novice) Cohort
Time 0	118	708	647 (91.4 %)	67.8 %	32.2 %
Time 5	114	684	570 (83.3 %)	61.4 %	38.6 %
Time 15	110	660	531 (80.5 %)	48.2 %	51.8 %
Time 30	110	660	522 (79.1 %)	50.9 %	49.1 %
Time 45	97	582	444 (76.3 %)	50.5 %	49.5 %
Time 60	96	576	455 (79.0 %)	52.1 %	47.9 %
Average				55.2 %	44.8 %

55.2 % plate assessments, compared to 44.8 % from their counterparts. Of the 645 plate inspections completed, the potential number of characters for interpretations was 3870 (6 characters per plate). As expected, not all characters could be deciphered by each participant and thus some results were left blank. This left a total of 3169 (81.9 %) characters identified, irrespective of accuracy. Fig. 3 shows the percentage of characters identified on each plate:

4.2. Error rate results

A total of 80 character errors were recorded in this study, after a total of 3169 recovered characters were assessed and identified. This provided a combined error rate of 2.52 % (80/3169). The distribution of character errors over the six time intervals is shown in Fig. 4 below.

The results of participants who produced errors were then isolated to determine any statistical outliers. To determine outliers, an interquartile range assessment was completed using the data of the 34 participants who contributed to the 80 misidentifications. The results are shown in the box and whisker graph (Fig. 5) below with the values above the maximum whisker line determined to be outliers. With the outlier results removed, the error rate for the expert cohort was adjusted to 2.4 %, and 2.3 % for the novice cohort.

4.3. Time 0 error

In forensic laboratories the examiner completing the serial number chemical etching is often the reporting officer, and thus a more accurate reflection of case-based error would be to exclusively look at the results of the participants that completed the serial number recoveries, the ‘Time 0’ participants. In this study 13 participants completed the chemical etching process (providing the Time 0 results). Due to the amount of time these participants had to view this process compared to other participants, it was expected that the Time 0 participants would provide less errors. The average error rate for these participants at Time 0 was determined to be 1.65 %. This error rate is significantly lower than their collective cohort averages at all the time intervals, and likely is a more accurate reflection of the error in casework. These results may demonstrate that serial number recovery is an evolving process, and that characters progress and fade over time.

4.4. Participant confidence results

Participant confidence scores (1–5 for each character identified) were then added and a total score of confidence for each plate assessment was determined. These total plate scores were grouped and the number of results per group shown, irrespective of time intervals, in

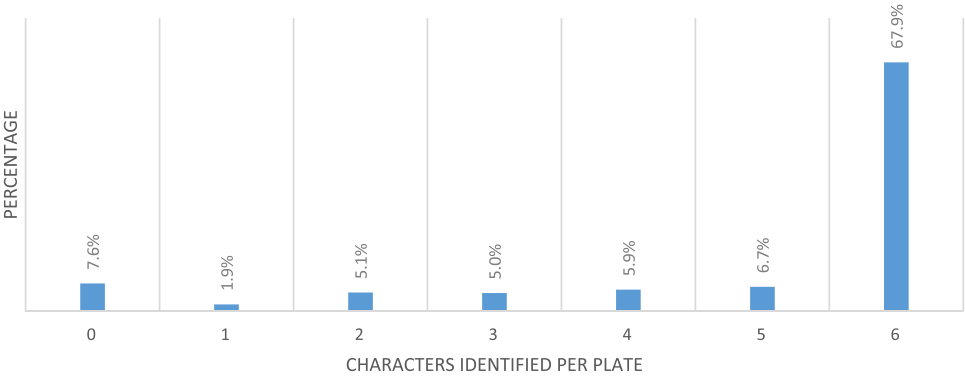


Fig. 3. Percentage of characters identified.

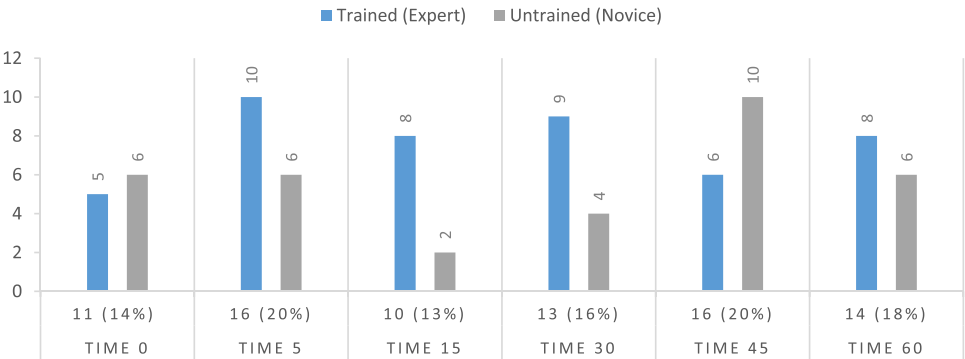


Fig. 4. Character errors over time.

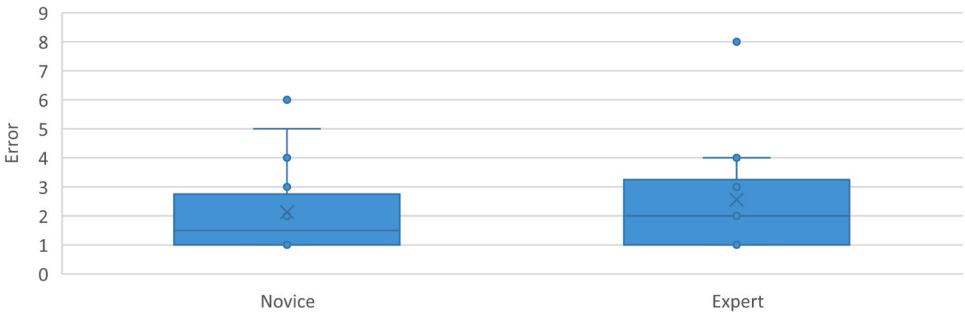


Fig. 5. Average Per Participant with Errors.

Table 2
Confidence results.

Cohort	Confidence Scores (Independent of time variable)					TOTAL
	6–10	11–15	16–20	21–25	26–30	
Trained (Expert)	63 (17.7 %)	43 (12.1 %)	64 (18.0 %)	85 (23.9 %)	101 (28.4 %)	356 (100 %)
Novice	57 (19.7 %)	26(9.0 %)	49(17.0 %)	49 (17.0 %)	108 (37.4 %)	289 (100 %)
TOTAL	120	69	113	134	209	645

Table 2. For example, the table shows a total of 63 incidences where an expert participant provided a confidence score of between 6 and 10 for their plate assessment.

To determine if there was a statistical difference between the expert and novice confidence scores, a Chi-Square test of Independence was completed, at a 95 % confidence. In chi-square test, the hypotheses were:

H₀: The cohort (trained/novice) and the confidence scores are independent,

H₁: The cohort (trained/novice) and the confidence scores are not independent.

The X² value was determined to be 9.5288, and a p-value of 0.04916. As the p-value is less than 0.05, the null hypothesis is rejected; the null hypothesis being the two cohorts are statistically similar in confidence.

Therefore a statistical difference exists between the cohorts with respect to their confidence scores. The novice group provided a confidence score of between 26 and 30 for 108 of the 289 results collected (37.4 %), compared to 101 of 356, (28.4 %), for the expert cohort. Based on the above and the Chi-Square test of independence results, it can be deduced that the novice group were more confident with their determinations, whilst the experts tended to be more conservative. Similar confidence assessment results have been found in studies relating to other forensic disciplines, such as fingerprint examination [23], and document examination [24].

The confidence scores were converted to a percentage and further assessed at the time intervals. Fig. 6 below shows the average confidence of both trained and novice groups at the six time intervals from recovery. As it is known that recovered markings can dissipate after time [25], it was expected that participant confidence would reduce as the time increased.

As illustrated by the two trend lines, the confidence of both participant cohorts decreased over time. The above results show a relatively consistent trend of novices being slightly more confident with their results compared to the trained (expert) group. Although the confidence of the groups inverts slightly at the 30 min mark, the trend lines do not overlap.

5. General discussion

There were a number of notable limitations placed on participants in this study, such as: preventing participants from utilising optical aids; using alternative chemical etchants; reapplying etchants; using alternate light sources or coloured lights; reparing the metals surface; and providing a maximum of 10 min for plate inspections. These restrictions were necessary to reduce assessment variables to provide more consistent results. Further studies can be completed which enables participants to draw on these resources in a controlled manner.

The etching procedure was conducted on each plate for a maximum on two hours. Based on previous literature and practitioner experience, two hours is generally considered to be a sufficient timeframe to undertake a serial number recovery. This is dependent on the chemicals used, the substrate material and the method of application. A total of 23 of the 118 plates in this study had their etching process terminated at the two hour mark. This was a larger number than expected and may allude to the difficulty of the restorations conducted. It is possible that some of these plates may have produced clearer results after this two hour threshold that may have led to less participant errors. As participants were also given limited time to assess each plate, it is therefore possible that time pressure was a factor that may have resulted in individuals translating characters incorrectly. Thus, these administrative errors may constitute a portion of the errors identified. It is not known if or how

often this occurred.

A preferred method of verification amongst jurisdictions is to have a second qualified examiner during the restoration to physically assess the etched surface. This allows a second independent examiner to form an opinion of the recovered characters which can later be discussed. The primary purposes of assessing results at the six time intervals was to determine if a time threshold could be identified, i.e., could a maximum time between recovery and verification be suggested based on the data collected. The results of this study showed that errors recorded at 5 min from recovery were just as prevalent at the 45 min mark. Thus, based on the time parameters tested, no conclusion can be drawn to determine a threshold for viewing and interpretation by a second assessor. Although, as recovered markings are known to deteriorate, it would be judicious to have a second assessment completed as soon as practicable to verify the result.

Optical aids such as the use of a magnifying glass, microscope or other optics, which enlarge or enhance the restored markings were prevented from use. These aids can help visualise detail that may be missed by the human eye alone and are often available for forensic examiners in laboratories. It is likely that the use of these aids assists during visual assessments of restored markings, and thus reduces potential misidentification errors.

The participants in this study were exposed to one font type at an ultimately uniform size and stamping. Furthermore they were provided a reference font template sheet to assist with deciphering alphanumeric characters during inspections. This process was to emulate a laboratory restoration where the examiner would seek knowledge of known serial number fonts to assist with their assessment. It would be expected that this information would assist in reducing identification errors. Further studies could be completed which involve using different character fonts, die stamp types (e.g. blunt face dies), or withholding information of the character font from participants. Fonts involving more character features such as serifs that assist in distinguishing characters will consequently reduce misidentifications. In order to provide a more thorough assessment of error rates that encompass the variety of serial number restorations practices completed in laboratories, further studies need to be completed.

In the laboratory forensic examiners are often further aware of the formats of serial numbers they are attempting to recover. E.g. the examiner may have baseline knowledge that the manufacturer of the exhibit only utilised serial number formats of a two letter prefix followed by six numbers. The examiner may also be aware of specific characters not used by the manufacturer, such as the letters 'I', 'O' and 'Q'. The test serial numbers used in this study are likely a larger range of alphanumeric characters compared to those found on firearms and vehicles. It is known that certain manufacturers will use a limited set of characters as serial numbers, likely in an effort to reduce misidentifications. An

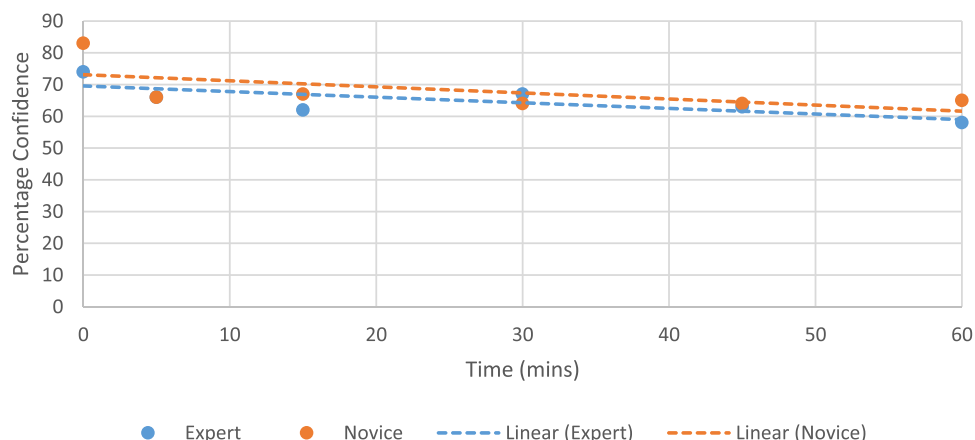


Fig. 6. Participant confidence per time interval.

examiner having this foundational knowledge will inherently prevent some misidentifications, thus reduce overall error rates.

As it is known that there is a large number of variables which can affect the quality, and therefore the interpretation of recovered characters, further studies need to be completed assessing the various aspects of serial number restorations. A large body of study has previously been completed in testing reagents and exploring new potential techniques for serial number recoveries. These studies however often involved a subset of alphanumeric characters and provided limited data on participant's prior knowledge of the pre-stamped characters. Although this studies have been beneficial for progressing the practice, this paper aimed to highlight other aspects of serial number restoration that need to be explored. Prior to this work, no research appears to have provided statistical data to represent the reliability of examiner interpretations of recovered serial numbers. This work aimed to assist in validating the interpretations of practitioners regularly undertaking restorations utilising a full set of alphanumeric characters as well as provide statistical data, through error rate and confidence analysis to further the knowledge of forensic examiners providing expert testimony.

As a whole, the number of character errors was found to increase over time whilst confidence of the participant decreases. An overall relatively low error rate was established for novice and expert cohorts (2.3 % and 2.4 % respectively). As all the participants in this study were English speaking, literate and regularly exposed to English alphanumeric, it was not surprising to see that there was a similar error rate between the two groups. Deciphering characters, e.g. reading a line of text, is done on a daily basis, however evidence of expertise may be seen in the comparison of confidence below. The novice group was found to be more confident with their results compared to the experts. Even only assessing the results with errors, the average confidence for the novice group was found to be 80 %, compared to 72 % for the expert cohort. This separation in confidence is consistent with expert-novice studies completed in other forensic disciplines.

This study focused at single assessor identifications of recovered characters, in both trained and untrained participants. General laboratory practices often involve a two-step verification process as a mitigation strategy to reduce potential errors. Thus a further evaluation of error rates is recommended that would encompass a full workflow assessment of serial number restorations, including the evaluation by a second assessor. It is hoped that this field of research is expanded on in future studies.

Although not covered as part of this work, expertise in serial number restoration may reside in the technical ability to prepare and chemically etch a materials surface. Participants which were part of the expert cohort had knowledge of serial number production, including the influence of die stamping and a range of avenues which can be used to recover obliterated markings. Careful consideration of an item is completed prior to undertaking surface preparation and serial number recovery. Different methods may be employed based on the location of the marking, its application method, and the surface deformations from the serial number removal process.

6. Conclusion

Serial number restoration is a practice that continues within forensic laboratories, potentially identifying important leads for investigators. Thus, it is crucial that the information derived from defaced items such as firearms and vehicles is reported accurately. This study aimed to provide statistical data to the reliability of the interpretations from chemically recovered characters.

To assist in validating practitioner interpretations, and addressing some forensic science criticisms, an error rate was calculated from the participant results. Taking into account the restrictions placed on participants for this study, low error rates were identified. The overall results showed a consistent accuracy between the expert and novice groups, but a statistical difference in confidence levels, demonstrating

the more conservative nature of experts. A total of 3169 character inspections were completed in this study. With the outlier results removed, an error rate of 2.4 % was determined for the expert cohort, and an error of 2.3 % for the novice group. When the results of the participants specifically undertaking the restorations were isolated, the average error rate was found to be only 1.65 %.

Prior to this work, the reliability of examiner interpretations of recovered characters had not been assessed. This research aimed to assist in validating the results of practitioners regularly undertaking restorations through error rate and confidence analysis, to better inform forensic examiners when providing results to court.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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